(Symmetric)

Bilinear Forms (Not in textbook)

Recall from Section 6.7 that an inner product on a <u>real</u> vector space satisfies

(i) $\langle u,v\rangle = \langle v,u\rangle \quad \forall\, u,v \in V$

(ii) $\langle u+v,w\rangle = \langle u,w\rangle + \langle v,w\rangle \quad \forall\, u,v,w \in V$

(iii) $\langle cu,v\rangle = c\langle u,v\rangle \quad \forall\, c\in\mathbb{R}$

(iv) $\langle u,u\rangle \geq 0$ and $\langle u,u\rangle = 0 \iff u=0$.

Def A <u>symmetric bilinear form</u> on a real vector space V is a function $B: V\times V \to \mathbb{R}$ satisfying,

(i) $B(u,v) = B(v,u)$

(ii) $B(u+v,w) = B(u,w) + B(v,w)$

(iii) $B(cu,v) = c\,B(u,v) \quad \forall\, c\in\mathbb{R},\ \forall\, u,v,w\in V.$

Example.

Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in M_{2\times 2}$ and define

$$B\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} u_1 & u_2 \end{bmatrix} A \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

i.e. $B(u,v) = u^T A v \quad \forall\, u,v\in\mathbb{R}^2.$

$$B\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \end{bmatrix}$$

$$= u_1 v_1 + u_2 0$$

$$= u_1 v_1$$

We will see later that B is a symmetric bilinear form.

To see that B is <u>not</u> an inner product lets compute

$$B\left(\begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right) = \cdots = (0)(0) = 0.$$

Clearly $u = \begin{bmatrix}0\\1\end{bmatrix}$ is not the zero vector, but

$B\left(\begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right) = 0$ so property (iv) fails in the definition of inner product.

Theorem.

Suppose B is a symmetric bilinear form on V. Then

$B(w, u+v) = B(w,u) + B(w,v)$ and

$B(u, cv) = c\,B(u,v) \quad \forall c \in \mathbb{R}$ and $u, v, w \in V$.

Sketch of proof.

$$B(w, u+v) = B(u+v, w) = B(u,w) + B(v,w) = B(w,u) + B(w,v)$$

Remark. The definition of symmetric bilinear form together with the theorem tell us that B is linear in each "slot". This is why we call it bilinear.

Theorem

Suppose B is a symmetric bilinear form on $\mathbb{R}^n$. Then there exists a symmetric matrix such that

$$B(u,v) = u^T A v.$$

Conversely if $A \in M_{n\times n}$ is symmetric then

$$B(u,v) = u^T A v$$ defines a symmetric bilinear form.

Sketch of proof.

If B is a symmetric bilinear form we can an matrix A whose jk entry is $a_{jk} = B(e_j, e_k)$ where e_j, e_k are standard basis vectors for $\mathbb{R}^n$.

Since $a_{jk} = B(e_j, e_k) = B(e_k, e_j) = a_{kj}$

we have $A^T = A$.

It turns out that $B(u,v) = u^T A v$, by expressing both u and v as linear combinations of the basis $\{e_1, \dots, e_n\}$.

Conversely suppose $A \in M_{n\times n}$ is symmetric. Define $B(u,v) = u^T A v$. We compute

$$\begin{aligned} B(u,v) &= u^T A v \\ &= (u^T A v)^T \\ &= v^T (u^T A)^T \end{aligned}$$

transpose of 1×1 matrix

$$= v^T A^T (u^T)^T \qquad A^T = A.$$

$$= v^T A u$$

$$= B(v,u)$$

For linearity, we compute

$$B(u+v,w) = (u+v)^T A w$$
$$= (u^T + v^T) A w$$
$$= u^T A w + v^T A w$$
$$= B(u,w) + B(v,w)$$

$$B(cu,v) = (cu)^T A v$$
$$= c\, u^T A v$$
$$= c\, B(u,v)$$

$A \in M_{n \times n}$ symmetric matrix

$B(u,v) = u^T A v$

symmetric bilinear form

$Q(v) = v^T A v$

Quadratic form

Example

Take $A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$, a symmetric matrix.

The bilinear form defined by A is given by

$$B\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 2v_1 + v_2 \\ v_1 - 3v_2 \end{bmatrix}$$

$$= u_1(2v_1 + v_2) + u_2(v_1 - 3v_2)$$

$$= 2u_1v_1 + u_1v_2 + u_2v_1 - 3u_2v_2$$

The quadratic form is given by

$$Q(v) = B(v,v) = v^T A v$$

$$= 2v_1^2 + 2v_1v_2 - 3v_2^2.$$

Q. When is a symmetric bilinear form B an inner product?

A. When $B(v,v) \geq 0$ and $B(v,v) = 0 \iff v = 0$.

We can rewrite this as $Q(v) \geq 0$ and $Q(v) = 0 \iff v = 0$.

where Q is the corresponding quadratic form.

This condition is equivalent to Q being positive definite. By Thm 5 §7.2 Q is positive definite $\iff$ its symmetric matrix has all positive eigenvalues.